ChatGPT

Continual Learning and Model Editing for Foundation Models: A Survey

Abstract: Foundation models – large-scale pretrained models in language, vision, and other modalities – have revolutionized AI by enabling transfer to diverse tasks. However, their static nature limits adaptability: they cannot easily incorporate new knowledge or correct errors without expensive retraining. Continual learning (CL) and model editing address these limitations from complementary perspectives. CL methods allow a model to update itself over a stream of tasks or data, mitigating catastrophic forgetting, while model editing techniques precisely inject or correct specific knowledge or behaviors in a model's parameters or inference process. This survey reviews recent advances at the intersection of these topics for foundation models. We first introduce the scope of continual adaptation and editing in large models, highlighting the need to maintain up-to-date knowledge, prevent forgetting, and ensure consistency when modifying models. In the **Methods** section, we categorize the literature into coherent families: for CL we discuss continual pre-training, continual fine-tuning (including parameter-efficient adaptation, rehearsal and regularization approaches, and prompt-based schemes), and evaluation protocols; for editing we cover fine-tuning and gradient-based edits, additional-parameter methods (hypernetworks, adapters, low-rank updates), "locate-and-edit" techniques (e.g. ROME/MEMIT), memory- or retrieval-augmented editing, and multi-modal editing. We describe key algorithms, assumptions, and performance trade-offs in each category. In **Challenges and Prospects**, we analyze open issues such as catastrophic forgetting, the stability–plasticity trade-off, efficiency and scalability constraints, evaluation gaps, and risks (e.g. degraded capabilities after edits, multi-turn editing conflicts). We also highlight future directions like continual compositional adaptation, reliable sequential editing, and standard benchmarks. This comprehensive survey aims to provide researchers with a coherent overview of how foundation models can be continually refined and corrected in a reliable, efficient manner, bridging CL and model editing literatures. **Keywords:** Continual learning, lifelong learning, model editing, foundation models, large language models, catastrophic forgetting, knowledge injection.

Introduction

Foundation models – large neural models pretrained on massive, broad-domain datasets – have become central to modern AI. Examples include transformer-based language models like GPT-3 and GPT-4, vision–language models like CLIP, and other large-scale pretrained networks【26 † L69-L72】【30 † L85-L94】. Such models serve as universal feature extractors or general-purpose predictors across tasks【26 † L69-L72】. Their impressive generalization, scale, and zero-shot capabilities have led to breakthroughs in natural language processing, computer vision, and beyond. However, a fixed pretrained model cannot easily adapt to new information or correct misunderstandings after deployment. Without retraining, a model's knowledge base remains static and can become **outdated** or **inconsistent** with evolving data and requirements【7 † L68-L74】【19 † L39-L47】. Moreover, naive full-model fine-tuning to adapt to new tasks can be prohibitively expensive and prone to catastrophic forgetting – the erasure of previously learned knowledge【61 † L31-L39】【64 † L37-L43】.

Two research areas address these challenges from different angles. **Continual learning** (CL), also called lifelong learning, studies how models can be trained over a sequence of tasks or data streams without forgetting earlier tasks【61 † L31-L39】【70 † L1-L4】. CL techniques (rehearsal, regularization, dynamic expansion, etc.) enable incremental updating of models on new tasks while retaining performance on past tasks. Many CL methods were developed before the rise of foundation models, but the unique scale

and capabilities of modern models raise new questions and opportunities in this domain【19 † L39-L47】【61 † L41-L49】. In parallel, **model editing** (or knowledge editing) focuses on precisely modifying a model’s behavior or knowledge after training. The goal is to inject, delete, or correct specific factual or behavioral outputs by changing model parameters or outputs with minimal side effects【64 † L37-L43】【50 † L30-L39】. Editing methods include fine-tuning on small sets of “edit examples,” adding small networks (hypernetworks, adapters) that adjust behavior, or even using retrieval and prompts to override model outputs at inference time. These techniques aim to update specific pieces of knowledge (e.g. correcting an incorrect fact the model produces) without harming unrelated capabilities.

In this survey, we focus on **Continual Learning and Model Editing for Foundation Models**. Our scope includes large pretrained models in language, vision, and multimodal domains. We consider how to incrementally update such models (via continued pretraining, fine-tuning, or prompting) and how to edit their knowledge or behaviors on-the-fly or post-training. We review and organize recent literature (2020–2026) covering both areas, identifying key methods, evaluation protocols, and empirical findings. Our aim is to provide a unified perspective: continual learning and model editing share the challenge of balancing **plasticity** (learning new information) and **stability** (preserving old knowledge), but approach it differently. Continual learning typically updates the entire model (often with memory of past data), whereas editing often seeks local or modular changes.

The remainder of this survey is organized as follows. In **Methods** (Sec. 2) we categorize approaches into coherent families and describe representative algorithms. In Sec. 2.1 we cover continual learning, from continual pre-training to continual fine-tuning (including parameter-efficient adaptation, replay, and prompt-based CL) and their assumptions. In Sec. 2.2 we address model editing, including fine-tuning or partial tuning, additional-parameter approaches (hypernetwork or adapter-based), locate-and-edit algorithms (e.g. ROME, MEMIT), and memory/retrieval-augmented methods (e.g. SERAC, RECIPE). We also highlight editing methods for multimodal models (BalancEdit, VisEdit) and language models. In **Challenges and Prospects** (Sec. 3), we discuss open problems such as catastrophic forgetting, evaluation difficulties, efficiency, and safety concerns (e.g. unintended generalization of edits【50 † L30-L39】). We conclude (Sec. 4) with a summary and outlook, emphasizing the need for systematic benchmarks and theoretical understanding. Throughout, we cite and synthesize findings from recent research to provide a comprehensive picture of how foundation models can be made continuously adaptable and correctable.

Methods

2.1 Continual Learning for Foundation Models

Continual learning (CL) methods seek to update models on a sequence of tasks or data streams while avoiding catastrophic forgetting【61 † L31-L39】. For foundation models, CL can occur at different stages. Three main paradigms have emerged: **Continual Pre-Training (CPT)** (also called “continued pretraining”), **Continual Fine-Tuning (CFT)** on task-specific data, and **Prompt- or Adapter-based CL** that treats the base model as frozen or black-box【61 † L47-L52】【54 † L13-L22】. In horizontal (temporal) CL, one learns across changing domains or time, while vertical CL adapts a general model to specialized skills【61 † L41-L49】.

- **Continual Pre-Training (CPT).** In CPT, one continues to train the foundation model on new raw data streams (unlabeled or self-supervised) to refresh its knowledge. Cossu et al. find that repeated self-supervised pretraining on incoming data substantially mitigates forgetting: a transformer continually pretrained on new NLP or vision data maintains prior knowledge better than standard CL techniques【29 † L89-L93】. This suggests that in large models, adapting

representations via more pretraining can itself preserve old knowledge. Key design choices for CPT include data selection (balancing old vs. new distributions) and learning rates【30 † L43-L52】【30 † L98-L105】. For example, Chazen et al. (2024) provide guidelines on schedule and data weighting for effective continued pretraining of LMs, showing that a mix of source and target data with carefully decayed learning rates can improve a 15B model’s general accuracy without collapse【30 † L43-L52】【30 † L98-L105】. CPT shares challenges with conventional CL, e.g. how to detect distribution shifts and avoid overfitting to new data.

- **Continual Fine-Tuning (CFT).** CFT refers to adapting a pretrained model to a series of downstream tasks, commonly by gradient updates. Traditional CL algorithms (replay buffers, regularization, parameter isolation) have been extended to large models. For instance, experience replay stores a small subset of past examples and interleaves them during fine-tuning to prevent forgetting【26 † L25-L34】【26 † L68-L77】. In vision, Ostapenko et al. study “latent replay”: they freeze a pretrained encoder and train only a classifier on top, either on raw data or on latent features. Surprisingly, simple latent replay (with a nonparametric classifier on frozen features) can yield reasonable CL performance when features are robust【26 † L25-L34】【26 † L78-L87】. They also show that broader pretraining (more data) improves CL transfer【26 † L69-L77】【26 † L78-L87】. Similarly, Momeni et al. propose a kernel-based CL method (KLDA) that represents each class in an LLM’s feature space by Gaussian distributions; this incremental classifier naturally adapts to new classes without fine-tuning model weights【46 † L7-L10】【47 † L483-L492】. Such approaches exploit frozen or partly frozen models for efficiency and privacy.

- **Parameter-Efficient CL.** Given the size of FMs, fine-tuning entire models for every task is costly. Recent work emphasizes parameter-efficient techniques: adapters, low-rank updates, hypernetworks or LoRA layers that add few trainable weights. For example, Wei et al. (2024) propose Online-LoRA, using LoRA on ViTs for online CL. They dynamically regularize LoRA weights and detect shifts via loss trends, achieving robust online learning without large buffers【42 † L28-L37】【42 † L38-L42】. Wang et al. (2024) introduce SEMA, which manages mixtures of adapter modules. They trigger new adapter addition only upon detecting a significant domain shift, and use a gating mechanism to route between old and new adapters. This allows sub-linear growth in parameters while reusing knowledge【44 † L39-L48】. More broadly, many CL methods now combine dynamic expansion (adding modules per task) with gating or routing to keep the model from growing unboundedly【56 † L125-L134】【56 † L138-L146】.

- **Prompt- and Black-Box CL.** Some works treat the FM as a black box and adapt via prompts or external memory. Qiu et al. (COLING 2025) introduce CLOB, a paradigm of CL over black-box LLM APIs using only verbal prompts【54 † L13-L22】【54 † L25-L34】. They use incremental summarization prompts (CIS) to encode class knowledge and update it as new examples arrive, thus avoiding any parameter updates. This reveals a prompt-based forgetting (CLOB) distinct from standard CL. Other works use prompt-tuning or prefix-tuning modules that grow or adapt over tasks. These strategies avoid touching core weights, which is appealing for proprietary models. They must, however, manage limited prompt context and ensure earlier knowledge summaries remain accessible.

- **Rehearsal and Hybrid CL.** Traditional CL techniques (experience replay, knowledge distillation, regularization like EWC/LwF) have been applied in FM contexts too. For instance, replay buffers combined with distillation losses have shown success in preventing forgetting in continual fine-tuning of LMs【30 † L119-L128】【30 † L129-L137】. In practice, many systems use a mix of parameter-efficient updates and selective replay. CL benchmarks for FMs are emerging (e.g., TiC-LM multi-year LLM pretraining benchmark, Chung et al., NeurIPS 2024) to systematically evaluate such methods on long streams.

2.2 Model Editing Techniques for Foundation Models

While CL adapts the entire model continuously, **model editing** aims to make targeted corrections or updates. Typical scenario: we have a trained FM that exhibits an undesirable behavior or outdated fact; instead of full retraining, we want to "patch" the model so that, say, "Paris is the capital of France" is correctly answered if previously incorrect. Key desiderata include specificity (the edit should affect only the intended knowledge, not unrelated outputs) and persistence (the edit should generalize to all relevant contexts)【64 † L37-L43】【50 † L30-L39】. We classify editing methods by how they introduce changes:

- **Fine-tuning / Gradient-based edits.** The simplest approach is to collect some edit examples (input-output pairs reflecting the corrected knowledge) and perform fine-tuning on those. While straightforward, naive fine-tuning often overfits or catastrophically disrupts the model. Techniques like **editing regularization** constrain changes. Early work like Sinitsin et al. (2020) introduced Knowledge Neurons by finding and updating specific weights, while Chen et al. (2023) show that even deleting some neurons can adjust outputs. Huang et al. (2023) propose T-Patcher, training a small network to generate a low-rank update that patches the model, as a more stable fine-tuning variant. We denote "FT" methods (full fine-tuning) and "FT-L" (fine-tune last layer only) in comparisons. The editing survey by Yao et al. (2023) groups these as brute-force editing【13 † L228-L239】.

- **Additional-Parameter Methods (Meta-learning, Hypernetworks, Adapters).** Instead of modifying base weights directly, these methods learn small auxiliary networks that generate edits. For example, MEND (Mitchell et al., 2022) and KE (De Cao et al., 2021) train a meta-network on many synthetic edits to produce weight updates via gradients; they rapidly propose updates at test time. SERAC (Mitchell et al., 2023) uses a small network to modify activations. T-Patcher (Huang et al., 2023) is similar. Another line uses adapters or low-rank layers: e.g., EMAD (Belinka et al., 2024) and NICAR (Gupta et al., 2023) train a compact patch network via rank-constrained matrices. LoRA-style updates can also serve as patches. The key idea is to confine edits to extra parameters, leaving the original model intact. This often improves locality (less interference)【33 † L278-L287】.

- **Locate-and-Edit (Neuron/Weight Surgery).** Some methods analyze the model to find which internal components "hold" the target knowledge. For example, ROME (Meng et al., 2022) identifies a rank-1 subspace in a transformer that encodes a specific fact (via automatic differentiation) and surgically updates it to inject a new fact. MEMIT (Zhu et al., 2023) iteratively extends ROME to multiple heads and layers to ensure broader effect. These methods are sensitive to model internals but can achieve very local edits. VisEdit (Chen et al., AAAI 2025) extends this idea to vision–language models: by attribution, it finds important visual features for a VLM's output and modifies intermediate visual representations to correct knowledge【38 † L53-L60】. Such approaches reveal the "mechanisms" by which models store facts, but they often assume white-box access and linear or low-rank structure in representations.

- **Retrieval- and Prompt-based Editing.** Recent work leverages retrieval or prompting at inference time to implement edits without altering weights. ICE (Xie et al., 2024) and RECIPE (Chen et al., EMNLP 2024) augment an LLM with retrieved "updated information" in the prompt. RECIPE, for instance, converts each new knowledge statement into a learned continuous prompt prefix, and prepends it to the query if the retriever finds a match【33 † L278-L287】【35 † L39-L47】. This allows the LLM to answer queries using the new knowledge as context. Because it does not change model parameters, this approach avoids long-term forgetting but requires a reliable

retrieval mechanism and may not fully “bake” the edit into the model. It can be viewed as an on-the-fly, input-side editing mechanism.

- **Memory-based Model Editing.** Some hybrid methods explicitly store edit data in an external memory. For instance, SERAC (Mitchell et al., 2023) trains an editor network to add residual memories to attention values, effectively revising model outputs. Memo (Lu et al., 2023) creates a memory bank of key-value pairs of edits. These systems blend retrieval with parametric editing.

- **Multi-modal and Vision-Language Editing.** Most editing research has focused on pure LMs, but foundation models span modalities. BalancEdit (Lee et al., ICML 2025) studies editing a multimodal model (vision+text) with one method: they train a small adapter for both text and image queries, balancing locality (not harming unrelated tasks) and generality (the edit works across modalities)【7 † L68-L74】. VisEdit (Chen et al., AAAI 2025) specifically targets VLMs, editing visual features as noted above【38 † L51-L59】. These works highlight that editing in multimodal models must account for how knowledge is distributed across modalities.

In summary, model editing methods fall into overlapping families: (a) straightforward fine-tuning; (b) meta-learned or auxiliary parameter methods (MEND, KE, T-Patcher, adapters/LoRA); (c) analytic locate-and-edit techniques (ROME/MEMIT, VisEdit); (d) retrieval/prompt-based patches (ICE, RECIPE, SERAC). Each makes different trade-offs in scope of edit, reliability, and practicality. Table 1 (not included here) in Yao et al. provides a comparison of representative methods on edit success, scope, and generalization【33 † L301-L309】.

Challenges and Prospects

Despite progress, continual learning and model editing face significant challenges when applied to foundation models:

- **Catastrophic Forgetting and Stability–Plasticity:** Even giant pretrained models suffer forgetting. CL methods must balance retaining core capabilities (stability) while integrating new data (plasticity)【61 † L31-L39】【29 † L89-L93】. In continual fine-tuning, distribution shifts across tasks can drastically hurt old-task performance, especially if the model overly specializes. For editing, Yang et al. (2024) show a “butterfly effect”: even a few targeted edits can collapse an LLM’s performance on other tasks【50 † L30-L39】. This indicates that edits, if not carefully constrained, can introduce fragility. A key open problem is designing edits (or CL updates) that provably preserve the model’s existing knowledge. Techniques like uncertainty-aware regularization, edit verification (e.g. monitoring perplexity or benchmark performance【50 † L30-L39】), and incremental prompts may help, but a general theory is lacking.

- **Evaluation Protocols and Benchmarks:** There is no unified standard for evaluating CL or editing on FMs. CL evaluations vary widely (different tasks, buffer sizes, metrics). Prominent CL benchmarks (PermutedMNIST, Split CIFAR) do not reflect the scale of modern models. Some new benchmarks are emerging: TiC-LM evaluates multi-year LM pretraining, CLoG (ICML workshop) targets generative model CL. For editing, tasks like CounterFact or MQUAKE provide single-shot edit tests; Yang et al. (HardEdit dataset) focus on worst-case failures【50 † L39-L47】. A comprehensive suite of editing benchmarks (for sequential edits, multi-turn, multi-modal edits) is needed. Moreover, reliable metrics are debated: edit success, locality (unintended changes), generalization, and training efficiency all matter.

- **Computational and Memory Efficiency:** Foundation models are large, so CL and editing must be efficient. Many works propose parameter-efficient updates, but even adapters/LoRA can accumulate parameters linearly with tasks. Methods like SEMA (adaptive adapter growth) and Online-LoRA (regularized LoRA) help, but memory growth remains a concern. Editing with auxiliary networks (MEND, SERAC) requires training these networks and storing them. The inference overhead of retrieval-based edits can be significant. Balancing efficiency with the thoroughness of updates is a hard tradeoff.

- **Scalability and Continuality:** In practice, foundation models are updated infrequently. True lifelong learning (open-ended stream of diverse, interleaved tasks) remains elusive. Many studies still assume clear task boundaries or limited domains. Handling task-free or streaming scenarios at FM scale (like online continual learning of LLMs across languages or modalities) is largely unexplored. The composition of models into multi-agent systems (continual compositionality) is a promising direction mentioned by Bell et al. (2025), but how to implement it concretely is open 【19 † L39-L47】.

- **Security, Bias, and Integrity:** Model editing introduces new risks. Malicious edits (poisoned updates) or adversarial use of editing mechanisms could damage a model or propagate misinformation. As Yao et al. discuss, edits should not introduce hallucinations or undesirable biases. The butterfly effect work warns that even well-intentioned edits can unexpectedly degrade capability【50 † L30-L39】. Ensuring robust, safe editing requires detecting when an edit "overflows" beyond its target and possibly rolling it back. Privacy is another concern: CL and editing often involve user data streams; methods like prompt-based CL mitigate direct parameter updates but could leak information via prompts.

- **Multimodal and Out-of-Distribution Robustness:** Foundation models now span vision, text, audio, etc. Techniques effective in one modality may fail in another. For example, editing visual grounding in VLMs (VisEdit) requires different methods than editing text facts. Handling continuous learning of capabilities like world modeling or reasoning is also open. Distribution shifts (e.g. cultural or temporal changes) affect foundation models profoundly. CL and editing must cope with out-of-distribution updates without destabilizing.

- **Future Directions:** Despite challenges, opportunities abound. Continual Compositionality: combining multiple specialized models (agents) via CL might yield systems that scale adaptively 【19 † L39-L47】. Meta-continual-learning: training FMs themselves to be better at future adaptation. Interactive Editing: allowing human feedback or reinforcement signals to guide sequential edits. Benchmarking and tools: as editing methods proliferate, standardized evaluation frameworks (like extension of CounterFact to multi-turn, multimodal) are needed. Finally, the intersection of CL and editing suggests hybrid methods: e.g., using small edit networks during CL, or treating edits as tasks in a lifelong learning framework.

Conclusion

Foundation models' extraordinary capabilities come with the responsibility and challenge of keeping them accurate and up-to-date. Continual learning and model editing are two key paradigms enabling this. In this survey, we reviewed how large pretrained models can be continually adapted via further pretraining, fine-tuning with memory, or prompt-based schemes, and how specific knowledge can be surgically edited using fine-tuning, hypernetworks, or retrieval. We organized methods into coherent categories and discussed representative algorithms and empirical findings. We also highlighted open problems: catastrophic forgetting in transformers, benchmarking difficulties, efficiency constraints, and

editing safety. As AI systems become integrated into dynamic real-world environments, solving these challenges is critical. We anticipate that ongoing research in continual adaptation, compositional intelligence, and robust editing will enable foundation models to evolve gracefully, incorporating new knowledge seamlessly and reliably.

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